Sheaf-cohomology QC for volume-EM registration: pre-registered negative results and a view-gauge theorem

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Abstract

We operationalized cohomological inconsistency (failure of adjacent-frame displacement measurements to form a 1-coboundary on the coverage nerve) as a quality-control detector for volume-electron-microscopy registration, and evaluated it, pre-registered, on three real channels: two distant ROIs of an EMPIAR-10478 HeLa SBF-SEM block and one vendor-registered EMPIAR-10070 FIB-SEM muscle stack. Three findings survive. (1) A **view-gauge theorem**: whole-view displacement is an exact gauge invariance of relative-content measurement; empirically confirmed — a shift-channel anomaly scores AUROC exactly 0.500 at every amplitude on real phase-correlation measurements. (2) **Minimal detectable bias is channel-physics-bound, not detector-bound**: an injected edge defect crosses AUROC 0.9 only at ∼4–5× the triangle-closure noise scale on raw SBF (GLR-optimal and max-abs detectors differ by <0.03), while pre-registered FIB-SEM residuals sit ∼15× lower in absolute pixels yet still admit no $\leq 3\times$ detection. The photogrammetry practice of pruning least-squares residuals above 2–3×RMS is falsified on all three channels, and closure-σ scales span two orders of magnitude across techniques: QC thresholds must be absolute per-task tolerances, never scale-free noise multiples. (3) An **instrument-disclosure protocol**: a self-referential-anchor defect that flat-lined the first detector at chance was caught by an algebraic gauge proof before any gate result was seen. All results, including nulls, are frozen in git before each run; code and artifacts are public at https://github.com/ScivizLabVienna/e3-sheaf-data.

1 Introduction

Large volume-EM reconstructions are assembled by registering thousands of adjacent section images; registration errors propagate silently into segmentation and connectomics. Existing automated quality assessment for this step is learned and pairwise — optical-flow SIQA metrics [4, 3], learned features [5], vision-language judges [6] — and carries no structural statement about *which* defects a given statistic can or cannot see.

Cellular sheaves on coverage nerves offer exactly such a statement: displacement measurements on overlaps are a 1-cochain, a consistent registration is a coboundary, and the closure residual of every triangle is a per-edge-usable inconsistency signal with no learned parameters. Prior registration work uses the same formalism to *eliminate* loop error globally (Poisson/Hodge solving) [2], never to *detect* per-edge defects against a noise-calibrated threshold.

We adopted the minimal instance of this detector, pre-registered acceptance gates, and ran it on real archives. The headline outcome is negative and useful: what the detector can see is set by channel physics, not by detector cleverness, and the two structural claims below are exact.

2 Formal setting

Frames $0, \ldots, N-1$ form a path nerve; each adjacent pair $(j, j+1)$ carries a measured translation $m_{jj+1} \in \mathbb{R}^2$ (phase correlation on overlapped ROIs). A *closure window* of K consecutive edges forms triangles whose residual is $\rho = m_{12} + m_{23} - m_{13}$ (global-reference legs measured independently at wide baseline). Two defect classes: **M_edge** — a wrong offset on one edge (e.g. a bad correspondence patch); **N_shift** — a whole-view translation desync.

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Theorem (view gauge). *Adding a per-view translation t_j to all measurements touching view j leaves every closure residual invariant; hence no coboundary-channel statistic can discriminate an N_shift at any amplitude.*

Proof: residuals are alternating sums in which each t_j appears with coefficients summing to zero. The empirical instance is exact, not approximate (§4).

Proposition (redundancy dilution). *In a window of K edges solved by weighted least squares, one corrupted edge's offset contributes to per-row residuals at $O(1/K)$; detection difficulty grows with the very redundancy that improves fit quality.*

3 Methods (frozen before each run)

Instrument (E6b): independent global-reference legs by wide-baseline (+30-frame) phase correlation plus adjacent lag-10 rows; triangle-closure residuals define σ_c per channel, emitted *before* any injection trial. Trials: $K = 6$, 40 trials per class, defect levels $\{0.5, 1, 2, 3, 5\} \times \sigma_c$; detectors max-abs residual and GLR-optimal whitened χ^2 energy [1]. Gate (identical for E6/E7/E8/E9): AUROC ≥ 0.9 at level $\leq 3 \times \sigma_c$, else channel-physics verdict. Channels: EMPIAR-10478 SBF ROI1 (576,297,858 B tile, 151×2000×2000), ROI2 (740,048,035 B), EMPIAR-10070 FIB-SEM MITOMAP stack [7] (777,443,899 B, 291×1595×1675, vendor TTrAC-registered). A synthetic campaign (P1–E5, oracle conditions) closed the space before real data.

4 Results

Table 1: Closure σ_c and GLR AUROC at $\{0.5, 1, 2, 3, 5\} \times \sigma_c$ (bootstrap CIs in artifacts). N_shift: 0.500 exactly at all levels (ROI1).

Channel	σ_c (px)	0.5×	1×	2×	3×	5×
SBF ROI1	1.78 / 1.66	0.600	0.720	0.844	0.890	0.943
SBF ROI2	1.46 / 1.47	0.593	0.696	0.823	0.881	0.927
FIB-SEM	0.11 / 0.14	0.571	0.586	0.628	0.666	0.737

No channel meets the gate at $\leq 3\times$; both SBF ROIs cross 0.9 only between 3× and 5× (E8 replicates E6/E7 on a distant ROI). GLR energy beats max-abs by $\leq +0.025$ AUROC everywhere: the curve is channel physics, not detector suboptimality. FIB-SEM inverts the mechanism — vendor pre-registration drives σ_c two orders of magnitude down, and redundancy dilution (Proposition) plus absolute-scale mismatch leave AUROC <0.9 even at 5×. *Closure-σ self-calibration does not normalize detection across channels.*

5 Discussion

Operational rule. The field practice of pruning registration residuals above 2–3×RMS [10] implicitly assumes noise-dominated channels; in deformation-dominated volume EM it both hides real defects (SBF) and flags benign sub-pixel residuals as unremarkable without information (FIB-SEM). Thresholds must be absolute per-task tolerances. The same caveat applies to calibration of learned QA metrics [4, 6].

Disclosure protocol. The first E6 instrument contained a self-referential-anchor defect that made the tested statistic an exact gauge by construction (AUROC flat 0.500). It was diagnosed algebraically and disclosed *before* any gate verdict; the rebuilt instrument (independent global legs) reproduced theory. We report this as a methodological contribution for agent-run science.

Limitations. One FIB-SEM stack (no ROI variety); no independent ground truth on EMPIAR-10070 (shifts baked by NEWSTACK; extended header byte-verified history-only), so the oracle-ceiling comparison was run only on SBF legs; serial-section TEM and array tomography untested.

6 Related work

Hodge/Poisson elimination of loop error in global registration [2]; EM stitching QA: EMSIQA [4], OFSIQA/DEMIS [3], learned features [5], VLM judges [6]; sheaf-inconsistency mathematics: consistency radius [8]; streaming H^1 [9]. None evaluates coboundary residuals as calibrated per-edge QC, and none reports pre-registered negative results for volume EM.

7 Reproducibility

Git-campaign record (every gate dated before its run), frozen evaluators, result artifacts and bootstrap CIs: `https://github.com/ScivizLabVienna/e3-sheaf-data`; source data by byte-size-verified EMPIAR accessions. Rebuild: `e6b_measure.py` → `e7_glr.py` (numpy/scipy/tifffile).

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